

```
# Student Management System
```

```
Students = {}
```

```
Action_stack = []
```

```
Waiting_queue = []
```

```
Def add_student():
```

```
    Roll = int(input("Enter Roll Number: "))
```

```
    Name = input("Enter Name: ")
```

```
    Mark = float(input("Enter Mark: "))
```

```
    Students[roll] = {
```

```
        "name": name,
```

```
        "mark": mark
```

```
    }
```

```
    Action_stack.append(f"Added {name}")
```

```
    Waiting_queue.append(name)
```

```
    Print("Student Added Successfully!")
```

```
Def view_students():
```

```
    If len(students) == 0:
```

```
        Print("No students available.")
```

Return

Print("\n--- Student List ---")

For roll, data in students.items():

Print(f"Roll No : {roll}")

Print(f"Name : {data['name']}")

Print(f"Mark : {data['mark']}")

Print("-" * 20)

Def search_student():

Roll = int(input("Enter Roll Number: "))

If roll in students:

Print("Student Found")

Print("Name :", students[roll]["name"])

Print("Mark :", students[roll]["mark"])

Else:

Print("Student Not Found")

Def delete_student():

Roll = int(input("Enter Roll Number: "))

If roll in students:

```
Name = students[roll]["name"]
```

```
Del students[roll]
```

```
Action_stack.append(f"Deleted {name}")
```

```
Print("Student Deleted Successfully")
```

```
Else:
```

```
Print("Student Not Found")
```

```
Def average_mark():
```

```
    If len(students) == 0:
```

```
        Print("No Students Available")
```

```
    Return
```

```
Total = 0
```

```
For data in students.values():
```

```
    Total += data["mark"]
```

```
Average = total / len(students)
```

```
Print("Average Mark =", average)
```

```
Def show_stack():
```

```
    Print("\nRecent Actions (Stack)")
```

```
If len(action_stack) == 0:
```

```
    Print("No Actions")
```

```
Else:
```

```
    For action in reversed(action_stack):
```

```
        Print(action)
```

```
Def show_queue():
```

```
    Print("\nWaiting Queue")
```

```
If len(waiting_queue) == 0:
```

```
    Print("Queue Empty")
```

```
Else:
```

```
    For student in waiting_queue:
```

```
        Print(student)
```

```
While True:
```

```
    Print("\n===== STUDENT MANAGEMENT SYSTEM =====")
```

```
    Print("1. Add Student")
```

```
    Print("2. View Students")
```

```
    Print("3. Search Student")
```

```
    Print("4. Delete Student")
```

```
    Print("5. Average Mark")
```

```
    Print("6. Show Stack")
```

```
Print("7. Show Queue")
```

```
Print("8. Exit")
```

```
Choice = input("Enter Your Choice: ")
```

```
If choice == "1":
```

```
    Add_student()
```

```
Elif choice == "2":
```

```
    View_students()
```

```
Elif choice == "3":
```

```
    Search_student()
```

```
Elif choice == "4":
```

```
    Delete_student()
```

```
Elif choice == "5":
```

```
    Average_mark()
```

```
Elif choice == "6":
```

```
    Show_stack()
```

```
Elif choice == "7":
```

```
    Show_queue()
```

Elif choice == "8":

Print("Thank You!")

Break

Else:

Print("Invalid Choice")